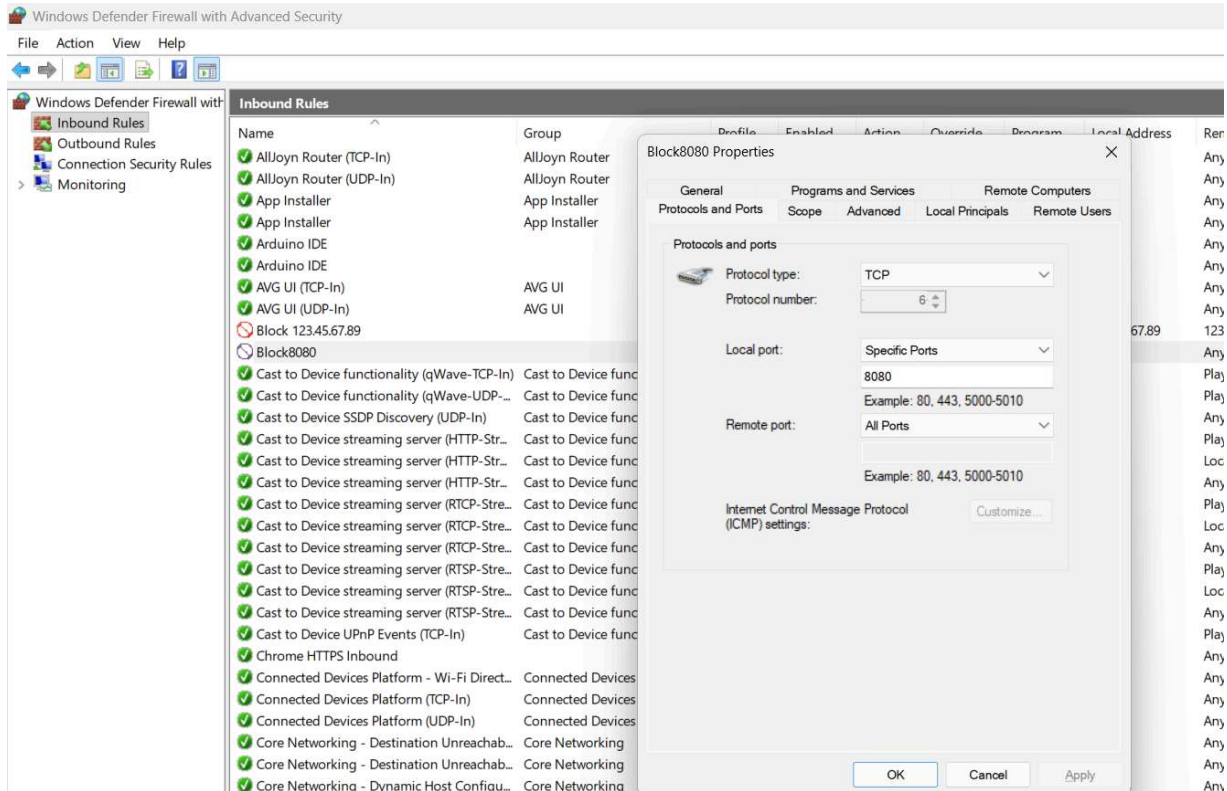
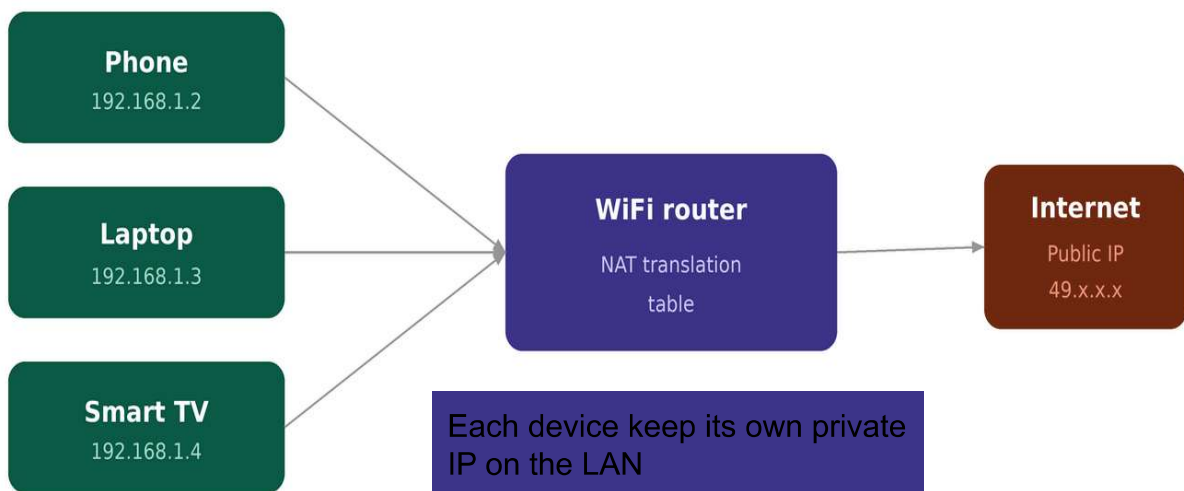


Task 1: Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.



Task 2: Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access. Hint: Use any drawing tool or even hand-draw and click a photo.



Router changes the source IP+assigns a port for each outgoing packet, then reserves it for replies (PAT)

The phone, laptop, and smart TV each will have a private IP (192.168.1.x) on the LAN, and the router's NAT table rewrites the source IP and assigns a port per connection so all three share one public IP going out so technically this is PAT — Port Address Translation, which is commonly used home-router.

Task 3: Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

Port 25 is the default network port used by the Simple Mail Transfer Protocol (SMTP) to transmit emails across the internet

Why Port 25 Is Mostly Blocked

In the early days of the internet, Port 25 transmitted traffic in unencrypted plain text and required no authentication. As a result, spammers and malware authors abused Port 25 to send massive volumes of unauthorized email.

To combat spam:

1. ISPs and Cloud Providers Block Port 25: Most residential Internet Service Providers (ISPs) and major cloud platforms (AWS, Google Cloud, Azure) block outbound traffic on Port 25 by default.
2. Email Clients Reject Plain Text: Modern email servers refuse unauthenticated email submissions on Port 25 from end-user devices.

In cisco Packet Tracer, we can use an **extended ACL** on a router instead of a "port filtering" panel, since Packet Tracer routers don't have a GUI firewall page—ACLs are how you do packet/port filtering there.

```
Router(config)#access-list 101 deny tcp 10.0.0.0 0.0.0.255 any eq 25
Router(config)#access-list ?
<1-99>      IP standard access list
<100-199>   IP extended access list
Router(config)#access-list 101 ?
deny        Specify packets to reject
permit      Specify packets to forward
remark      Access list entry comment
Router(config)#access-list 101 permit ?
ahp         Authentication Header Protocol
eigrp       Cisco's EIGRP routing protocol
esp         Encapsulation Security Payload
gre         Cisco's GRE tunneling
icmp        Internet Control Message Protocol
ip          Any Internet Protocol
ospf        OSPF routing protocol
tcp         Transmission Control Protocol
udp         User Datagram Protocol
Router(config)#access-list 101 permit ip ?
A.B.C.D     Source address
any         Any source host
host        A single source host
Router(config)#access-list 101 permit ip any ?
A.B.C.D     Destination address
any         Any destination host
host        A single destination host
Router(config)#access-list 101 permit ip any any ?
dscp        Match packets with given dscp value
precedence  Match packets with given precedence value
<cr>
Router(config)#access-list 101 permit ip any any
Router(config)#exit
```

10.0.0.0 0.0.0.255 → source network + wildcard (your LAN)

any → destination (any host)

eq 25 → destination port 25 (SMTP)—this only goes after the destination, since we are filtering the traffic *going to* port 25

The network diagram shows a central router (R1) connected to two switches (S1 and S2). S1 is connected to a PC (PC1) and a server (S1). S2 is connected to a PC (PC2) and a server (S2). The router has interfaces Fa0/0, Fa0/1, Fa0/2, and Fa0/3. The PC1 has IP 10.0.0.1 and the server S1 has IP 10.0.0.2. The PC2 has IP 10.0.0.1 and the server S2 has IP 10.0.0.2. The router has a loopback interface Lo0 with IP 10.0.0.1.

The PDU Information window is open, showing the details of a packet being sent from the router to the destination. The packet is a TCP packet with source IP 10.0.0.1 and destination IP 10.0.0.2. The window shows the packet's structure across seven layers, from the Physical layer to the Application layer. The packet is being sent from the router's Fa0/0 interface to the destination PC.

PDU Information at Device: Router 1 to ISP

OSI Model | Inbound PDU Details | Outbound PDU Details

At Device: Router 1 to ISP
Source: PC0
Destination: 30.0.0.3

In Layers

Layer 7
Layer 6
Layer 5
Layer 4
Layer 3: IP Header Src. IP: 10.0.0.2, Dest. IP: 30.0.0.3
Layer 2: Ethernet II Header 00E0:F705:EE5E >> 0002:1713:4901
Layer 1: Port GigabitEthernet0/0

Out Layers

Layer 7
Layer 6
Layer 5
Layer 4
Layer 3: IP Header Src. IP: 10.0.0.2, Dest. IP: 30.0.0.3
Layer 2
Layer 1

1. The routing table finds a routing entry to the destination IP address.
2. The device decrements the TTL on the packet.
3. The routing table finds the outgoing interface access-list with an ID of 101. The device checks the packet against the access-list.
4. The packet matches the criteria of the following statement: deny tcp 10.0.0.0 0.0.255 any eq smtp. The packet is denied and dropped.

Challenge Me | << Previous Layer | Next Layer >>

Note: default behavior of Cisco ACLs: **every ACL has an invisible "deny all" at the very end**, even though you never type it.

Adding: *access-list 101 permit ip any any*

explicitly tells the router, "For anything that didn't match the deny-port-25 rule above, allow it through normally." This restores normal connectivity for everything except SMTP.

PDU Information at Device: Router 1 to ISP

OSI Model

Inbound PDU Details

Outbound PDU Details

At Device: Router 1 to ISP
Source: PC0
Destination: 30.0.0.3

In Layers

Layer7

Layer6

Layer5

Layer4

Layer 3: IP Header Src. IP:
10.0.0.2, Dest. IP: 30.0.0.3

Layer 2: Ethernet II Header
00E0.F705.EE5E >>
0002.1713.4901

Layer 1: Port GigabitEthernet0/0

Out Layers

Layer7

Layer6

Layer5

Layer4

Layer 3: IP Header Src. IP:
10.0.0.2, Dest. IP: 30.0.0.3

Layer2

Layer1

1. The routing table finds a routing entry to the destination IP address.
2. The device decrements the TTL on the packet.
3. The outgoing port has an outbound traffic access-list with an ID of 101. The device checks the packet against the access-list.
4. The packet matches the criteria of the following statement: deny tcp 10.0.0.0 0.0.0.255 any eq smtp. The packet is denied and dropped.

Challenge Me

<< Previous Layer

Next Layer >>

Task 4 : Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53 Hint: Focus on source/destination IPs and ports in the rule set.

Rule ID	Action	Protocol	Source IP	Source Port	Destination IP	Destination Port	Description / Purpose
Rule 1	ALLOW	TCP	192.168.1.0/24	Any	Any (0.0.0.0/0)	80 (HTTP)	Allow local subnet web traffic
Rule 2	ALLOW	UDP	192.168.1.0/24	Any	8.8.8.8	53 (DNS)	Allow local subnet DNS queries to Google DNS
Rule 3	DENY	TCP	Any (0.0.0.0/0)	Any	Any (0.0.0.0/0)	22 (SSH)	Explicitly block SSH outbound access
Rule 4	DENY	Any	Any (0.0.0.0/0)	Any	Any (0.0.0.0/0)	Any	Default Implicit Deny

Packet 1: HTTP Request

- **Attributes:** 192.168.1.5 - 172.217.166.78:80 (TCP)
- **Verdict:** **ALLOWED**
- **Reasoning:** Matches **Rule 1**. The source IP 192.168.1.5 falls within the allowed subnet 192.168.1.0/24, and the destination port is 80 (HTTP).

Packet 2: SSH Request

- **Attributes:** 10.0.0.10 - 34.201.23.1:22 (TCP)
- **Verdict:** **BLOCKED**
- **Reasoning:** Matches **Rule 3** (and eventually **Rule 4**). SSH destination port 22 is explicitly prohibited across all IP ranges. Additionally, the source IP 10.0.0.10 does not belong to the permitted 192.168.1.0/24 range.

Packet 3: DNS Query

- **Attributes:** 192.168.1.7 - 8.8.8.8:53 (UDP)
- **Verdict:** **ALLOWED**
- **Reasoning:** Matches **Rule 2**. The source IP 192.168.1.7 is within 192.168.1.0/24, the destination target is explicitly 8.8.8.8, and the service port is 53 (DNS over UDP).

Task 5: Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

Rate limiting and DDoS protection at the edge Firewalls (often with a CDN/WAF layer such as Cloudflare or AWS Shield) identify and drop traffic when a single IP or botnet generates an abnormally high number of requests in a short period, e.g. during a bot-driven login-spam attack or a coordinated DDoS attempt. This prevents backend application servers from being overwhelmed with traffic that is not real users.

Malicious or known-bad IP range blocking - Packet filtering rules can automatically drop traffic from IP addresses or entire ranges associated with known botnets, scraper farms, or previous attack attempts before that traffic ever reaches the application layer. This is often combined with geo-blocking, e.g. blocking traffic from regions where the service does not have a valid user base.

Segment internal infrastructure - databases, internal admin panels, and payment-processing systems are behind internal firewalls in private subnets, and have strict rules that only allow specific application servers (not the public internet) to reach sensitive ports — e.g. a database port like 3306 (MySQL) or 5432 (PostgreSQL). This segmentation also limits the “blast radius” of an attacker, should he or she be able to compromise a public-facing web server, to more sensitive systems.